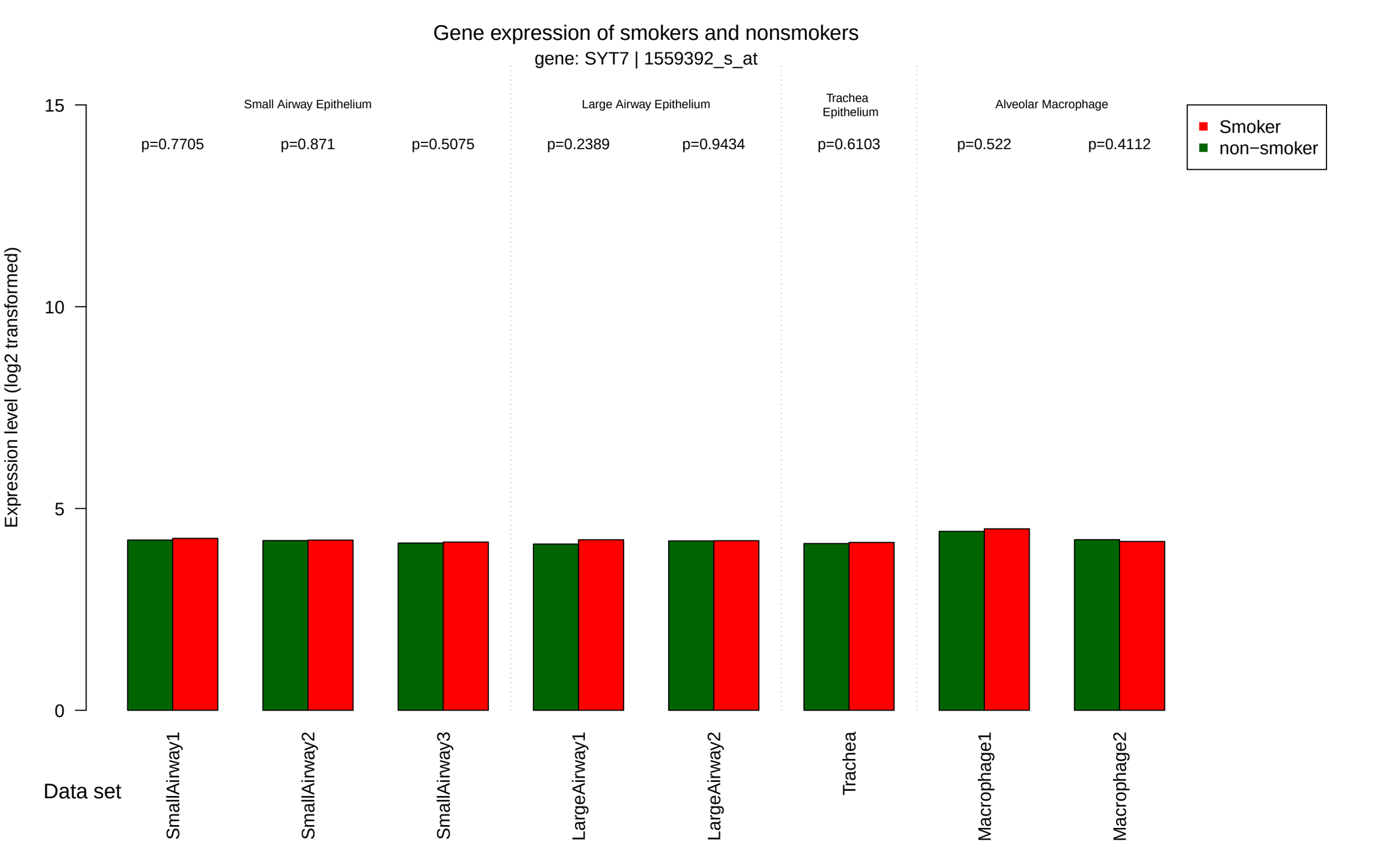
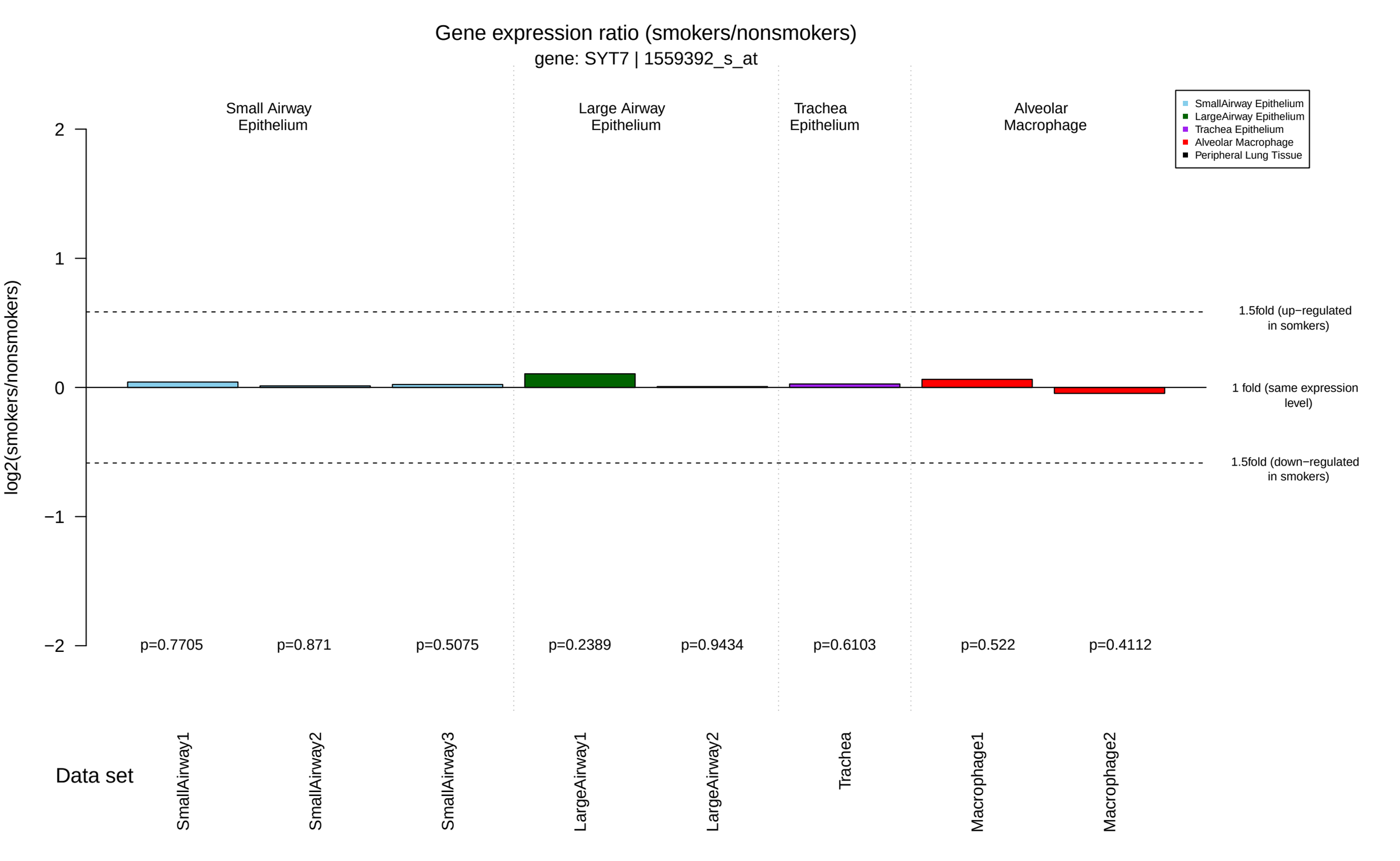
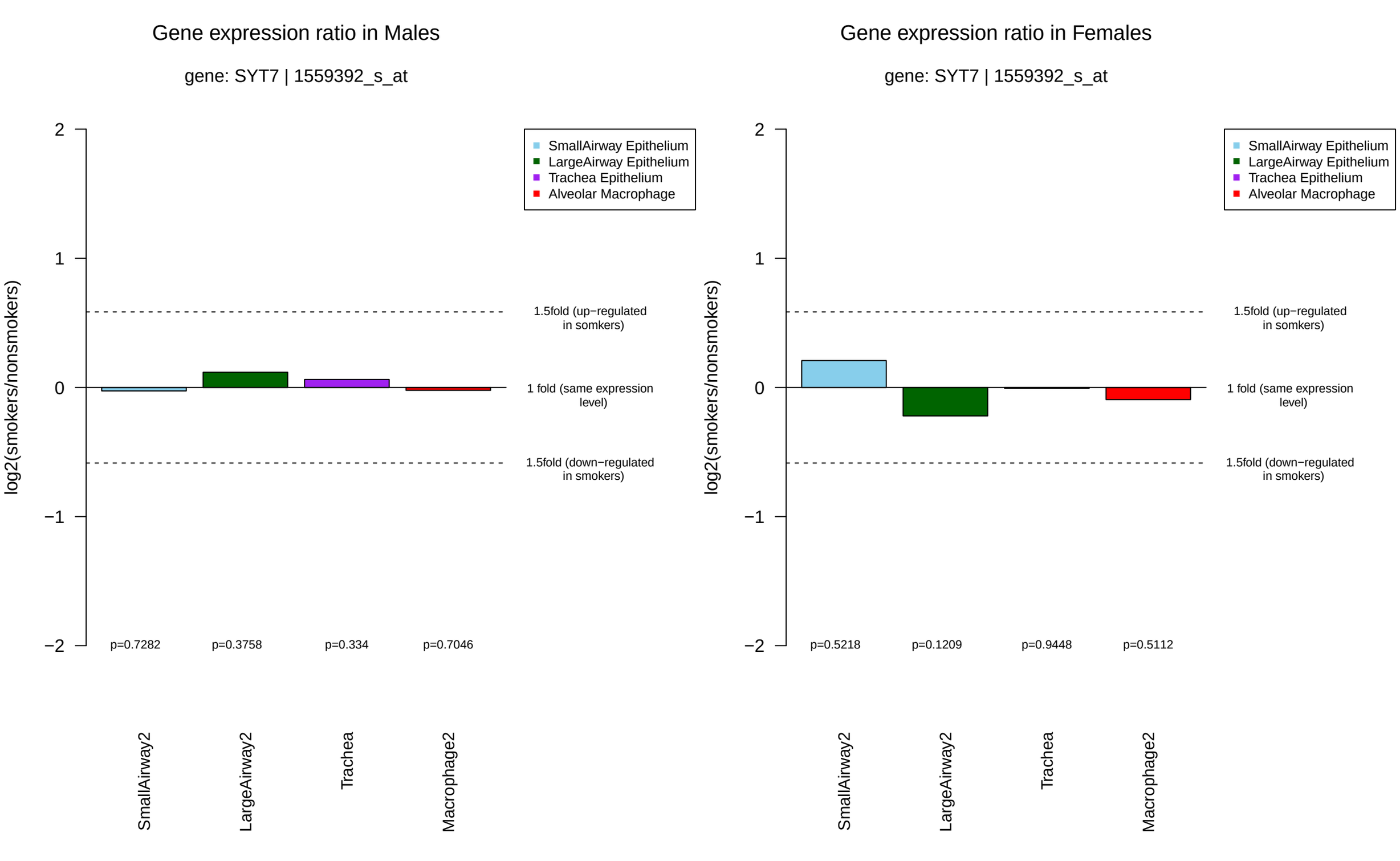
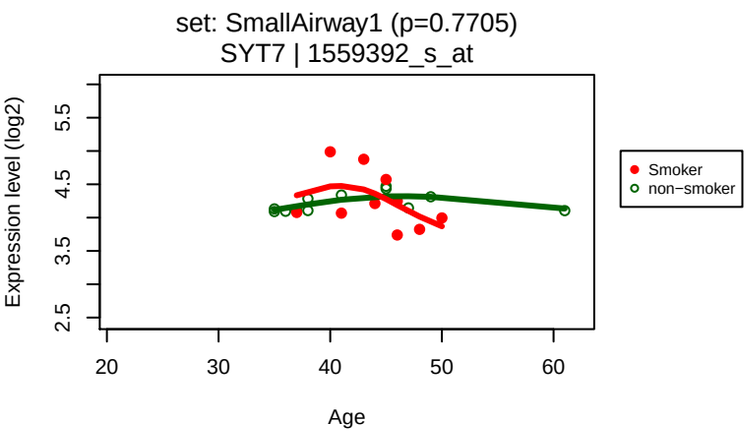
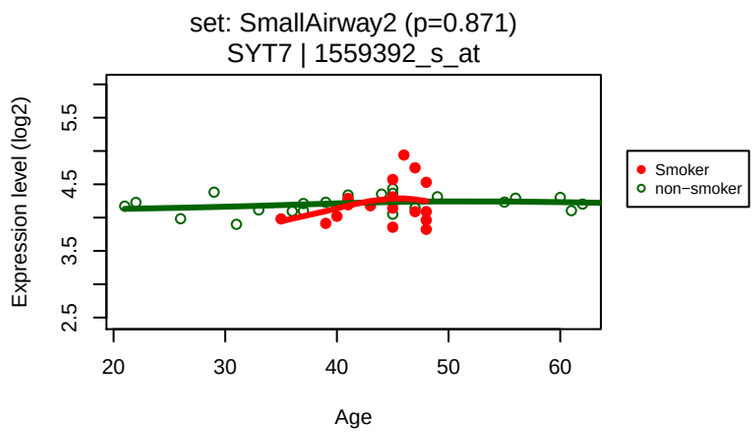
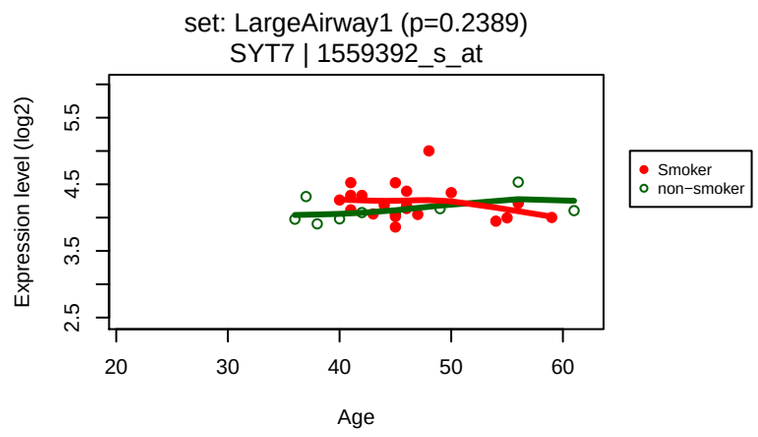

Gene expression of smokers and nonsmokers
gene: SYT7 | 1559392_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7705
p=0.871
p=0.5075
p=0.2389
p=0.9434
p=0.6103
p=0.522
p=0.4112
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

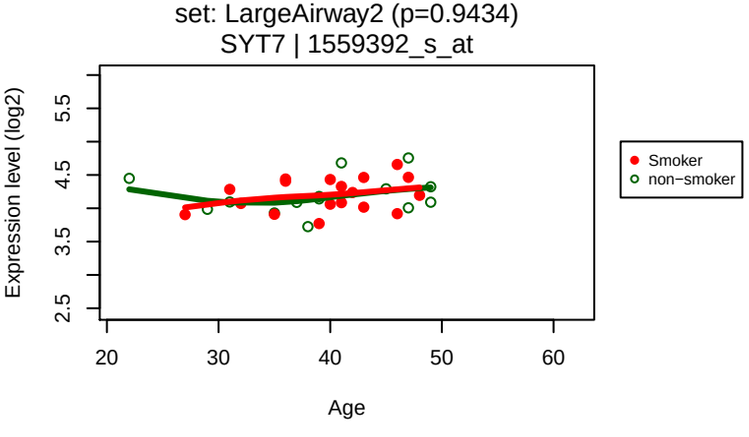

Gene expression ratio (smokers/nonsmokers)
gene: SYT7 | 1559392_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7705
p=0.871
p=0.5075
p=0.2389
p=0.9434
p=0.6103
p=0.522
p=0.4112
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

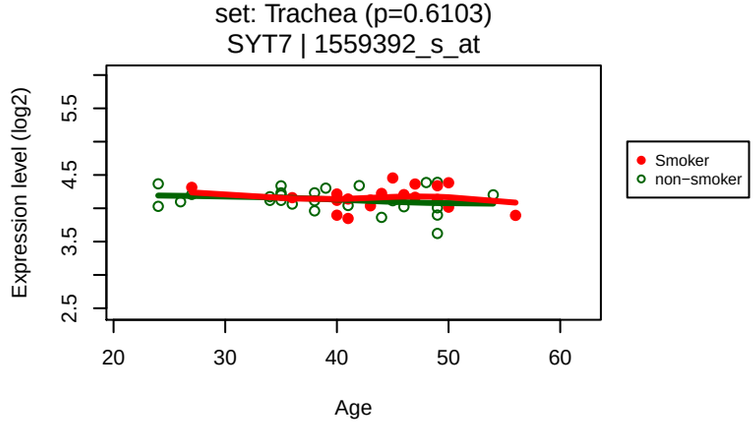

Gene expression ratio in Males
gene: SYT7 | 1559392_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7282
p=0.3758
p=0.334
p=0.7046
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SYT7 | 1559392_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5218
p=0.1209
p=0.9448
p=0.5112
SmallAirway2
LargeAirway2
Trachea
Macrophage2

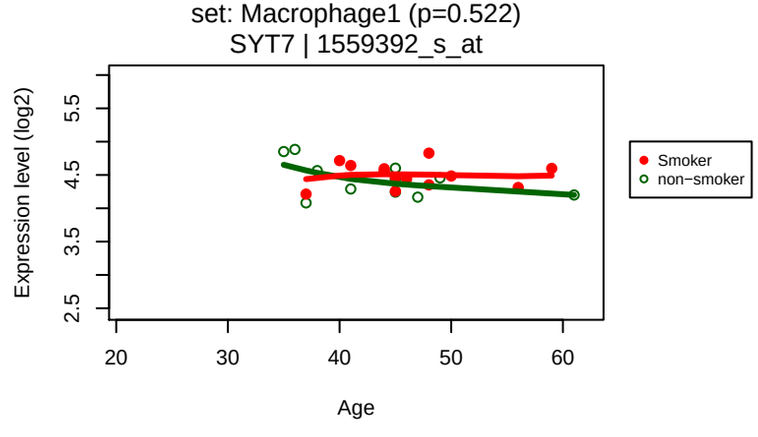

set: SmallAirway1 (p=0.7705)
SYT7 | 1559392_s_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker

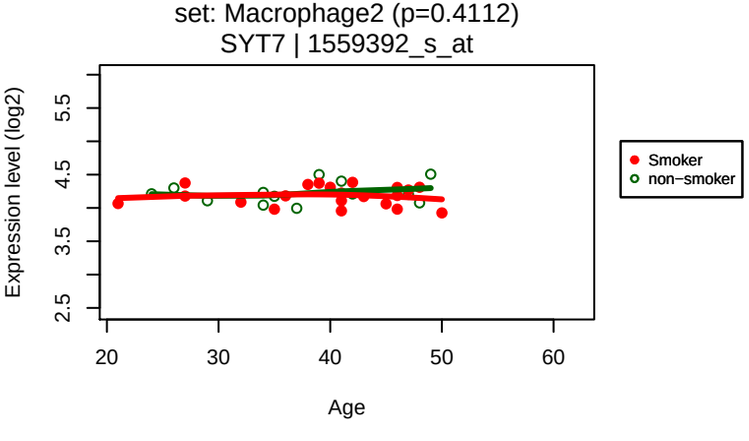

set: SmallAirway2 (p=0.871)
SYT7 | 1559392_s_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker

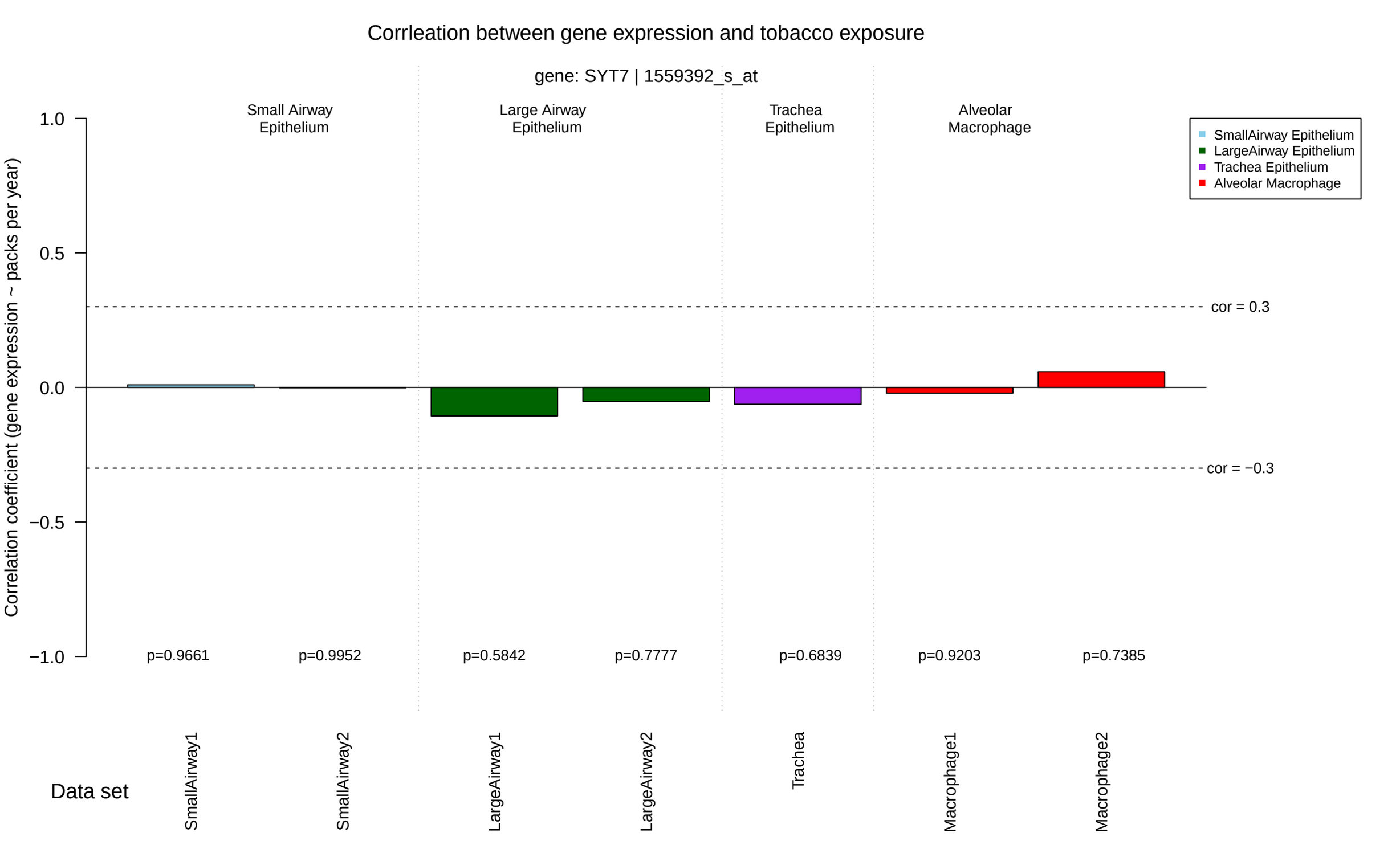

set: LargeAirway1 (p=0.2389)
SYT7 | 1559392_s_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9434)
SYT7 | 1559392_s_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.6103)
SYT7 | 1559392_s_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.522)
SYT7 | 1559392_s_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4112)
SYT7 | 1559392_s_at
Expression level (log2)
2.5
3.5
4.5
5.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SYT7 | 1559392_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9661
p=0.9952
p=0.5842
p=0.7777
p=0.6839
p=0.9203
p=0.7385
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2